**SIMATS ENGINEERING**

Department of Computer Science and Engineering

**DSA04 – Fundamentals of Data Science**

**ASSIGNMENT 1 – CO1**

**Data Science for Informed Decision-Making in a Smart University**

*CO1: Need for Data Science, benefits and uses, facets of data, Big Data ecosystem, Data Science process – retrieving, cleansing, integrating and transforming data, data analysis, model building, presenting findings and building applications*

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| **Department** | Computer Science and Engineering | **Academic Year** | 2024-2028 |
| **Faculty Name** | Dr. Vanaja Ramalingam | **Nature of Assignment** | Individual |
| **Duration** | 1 Week | **Total Marks** | 50 |
| **Date of Submission** | 02/11/2026 | **Date of Evaluation** | 03/11/2026 |

**Mark Distribution**

| Task | Task Title | Max | Marks Awarded |
|---|---|---|---|
| 1 | Need and Role of Data Science | 4 | |
| 2 | Benefits and Applications of Data Science | 4 | |
| 3 | Identify the Facets of Data | 5 | |
| 4 | Analyse the Characteristics of Big Data | 5 | |
| 5 | Big Data Ecosystem and Its Components | 5 | |
| 6 | Data Retrieval and Data Cleansing | 4 | |
| 7 | Data Integration and Transformation | 5 | |
| 8 | Data Analysis and Model-Building Stages | 5 | |
| 9 | Presenting Findings and Building Applications | 5 | |
| 10 | Integrated Data Science Decision-Making | 8 | |
| | **TOTAL** | **50** | |

| | | |
|---|---|---|
| **Student Signature** | **Faculty Signature** | **Program Director** |

## FUNDAMENTALS OF DATA SCIENCE

**Assignment Title: Data Science for Informed Decision-Making in a Smart University (Total Marks – 50)**

**Scenario – Smart University Decision-Support System.** A university wants to improve academic planning and student-support decisions using information generated through its **student information system, attendance records, examination system, learning management system, library system** and **online feedback platforms**. These sources contain data in **different formats** and are **growing continuously**. The university plans to use a **Data Science approach** to convert this data into useful findings for faculty members and administrators.

I am assigned as a member of the university's Data Science team. Tasks 1 to 10 below relate each concept to this scenario, with examples, tables, diagrams and workflow representations wherever requested.

### How this report is organised

Each task gives The question as set, followed by my answer under **Ans:**. Tables and figures are numbered in order. Task 5 contains the labelled Big Data ecosystem diagram and Task 10 contains the overall flowchart. Every number used in an example is either an **assumption** (listed below) or an **illustrative value** and is marked as such; none of them is a measured result from a real university.

### Assumptions made in this report

1. The university has about **15,000 students**, each studying about **6 courses** per semester, with **6 teaching periods per day**.
2. The six source systems exist as separate systems. Each student has a **student ID** in the student information system; the other systems store a student ID or register number that can be matched to it.
3. The LMS records about **150 click events per student per day** (login, page view, video play, quiz attempt, forum post), each of about **0.5 KB**. About **40 hours of lectures per day** are recorded, at about **0.7 GB per hour**.
4. Student data is used only for academic support and planning. Names are replaced by a coded key in the analysis dataset, and access is role-based.
5. Tools such as Hadoop, Spark, Kafka, NoSQL stores and dashboard software are named only as **examples** of a component. The answer does not depend on one product, and no machine-learning algorithm is implemented (as stated in Tasks 8 and 9).

### Relation to CO1

CO1 asks for an analysis of the role of Data Science, the facets of data and the Big Data ecosystem for informed decision-making. Tasks 1–2 give the need and the applications, Task 3 the facets of data, Tasks 4–5 the Big Data characteristics and ecosystem, Tasks 6–9

## TASK 1 – Need and Role of Data Science 4 Marks

Explain why Data Science is required in the Smart University scenario.

- State the need for Data Science in handling university information.
- Explain two benefits of using Data Science for academic or administrative decision-making.

**Evidence Expected**

A clear explanation of the need and role of Data Science with relevant university examples.

### 1. Why the Smart University needs Data Science

The university runs **six information systems**, and every one of them records something different about the same student. The student information system holds the profile, the attendance system holds presence, the examination system holds marks, the LMS holds study behaviour, the library holds borrowing, and the feedback platform holds opinions. Each system answers only its own question. The questions that management really asks – Which students are likely to fail? Why is participation falling? Where should we add support? – need **all of them together**, and need answers **early enough to act**. Data Science is the approach that makes this possible. The main reasons are:

- **The data is scattered in separate systems.** A faculty member can see a student's marks but not that the student stopped logging in to the LMS a month ago. Joining the sources gives one complete picture of the student.
- **The data is too large and too varied for manual methods.** Millions of LMS click events, thousands of feedback comments and daily attendance rows cannot be read or summed by hand or in a spreadsheet, and free-text comments cannot be tabulated at all without text-analysis methods.
- **Decisions are normally taken late.** Marks appear only after an examination, so a failing student is discovered when it is too late to help. Attendance and LMS data arrive every day and can show the warning signs weeks earlier.
- **Important patterns are hidden.** A fall in attendance, a drop in LMS logins and a late submission may each look harmless alone, but together they may be a strong sign of risk. Such combinations are found only by systematic analysis.
- **Decisions need evidence, not opinion.** Timetabling, laboratory allocation, library purchases and teaching improvements are expensive. Data-based findings let administrators justify and later check each decision.

**Table 1: Handling university information without and with Data Science**

| Aspect | Without Data Science | With Data Science |
|---|---|---|
| **Data view** | Each department looks at its own system | One combined view of each student and course |
| **Speed of decision** | After the result or the complaint | Weekly or daily, before the problem grows |
| **Type of question** | What happened? (reports) | Why did it happen and what is likely next? |
| **Data handled** | Neat tables only | Tables, logs, comments and files together |
| **Basis of decision** | Experience and opinion | Measured evidence that can be re-checked |

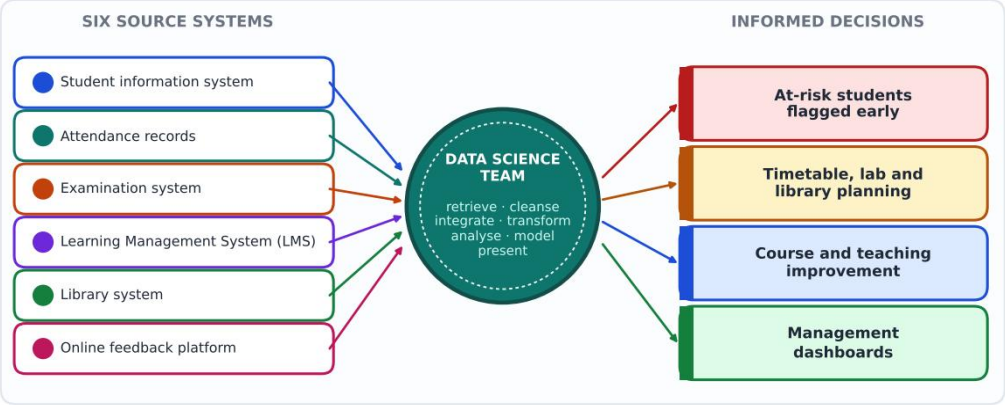



*Figure 1: Data from six university systems is turned into informed decisions by the Data Science team*

## 2. The role of Data Science

The role of Data Science is to **convert raw data into findings that decision-makers can act on**. In this scenario it works at four levels, each answering a more useful question than the one before:

**Table 2: Four levels of questions answered by Data Science**

| Level | Question answered | University example |
|---|---|---|
| **Descriptive** | What has happened? | Average attendance of each department last month |
| **Diagnostic** | Why did it happen? | Participation fell most in courses with long video lectures and poor feedback |
| **Predictive** | What is likely to happen? | Which students are likely to fall below the pass mark this semester |
| **Prescriptive** | What should we do? | Which support action should be given first to which group of students |

## TASK 2 – Benefits and Applications of Data Science 4 Marks

Identify and explain suitable applications of Data Science in a university environment.

- Describe at least four possible applications, such as student-support planning, resource planning, learning analysis or feedback analysis.
- Explain how any two applications can support informed decisions.

**Evidence Expected**

Relevant applications and a clear link between data-based findings and decisions.

### 1. Applications of Data Science in the university

The six data sources of the scenario support many applications. Table 3 lists six that are suitable, the data each one needs and the decision it supports. The first four are the ones named in the question; the last two show that the same data can serve more purposes.

**Table 3: Applications of Data Science in the Smart University**

| Application | What it does | Data used | Decision supported |
|---|---|---|---|
| **A1. Student-support planning** | Finds students likely to struggle and shows the reasons | Attendance, internal marks, LMS activity, submission record | Whom to counsel or coach, and when |
| **A2. Resource planning** | Measures use of classrooms, laboratories and library seats and forecasts demand | Timetable, room bookings, library entry and loan records | How to allocate rooms, lab hours and staff; whether to buy more |
| **A3. Learning analysis** | Studies how students use the course material and where they stop | LMS clicks, video views, quiz attempts, forum posts | Which content to shorten, redesign or add; when to schedule quizzes |
| **A4. Feedback analysis** | Reads comments and ratings to find common complaints and praise | Feedback ratings and free-text comments | Which courses or teaching methods need improvement |

### 2. How two applications support informed decisions

#### Application A1 – Student-support planning

**Data-based finding.** The team combines weekly attendance, internal-test marks, LMS logins and assignment submissions for each student and compares them with the pattern of earlier students who passed or failed. Students whose indicators fall below agreed limits are marked as *at risk*, together with the indicator that triggered the mark.

**Table 4: From finding to decision – student-support planning (illustrative)**

| Finding (illustrative) | Informed decision | Expected effect |
|---|---|---|
| 12 % of semester-5 students have attendance below 70 % and no LMS login for 3 weeks | Mentors meet these students in week 6; counsellor sessions are given priority | Problems are handled before the internal examinations |
| The at-risk share is 22 % in one course but 9 % in others | Course coordinator reviews the pace and difficulty of that course; extra tutorial hours are added | Support goes to the course that needs it |
| Students who submitted the first two assignments late are twice as likely to be at risk | Reminders and short check-ins are sent after the first late submission | Risk is spotted by the third week |

**Application A2 – Resource planning**

**Data-based finding.** The team joins the timetable with room-booking and access records to calculate how much of the available time each classroom and laboratory is really used, and compares this with the number of students in the sessions. Library entry and loan counts are analysed in the same way by week.

**Table 5: From finding to decision – resource planning (illustrative)**

| Finding (illustrative) | Informed decision | Expected effect |
|---|---|---|
| Laboratories L1 and L2 are used for 38 % of available hours; L7 and L8 for 96 % | Move two practical batches from L7 and L8 to L1 and L2 | Less overcrowding; no new laboratory needed |
| Library seat occupancy is above 90 % in the three weeks before examinations and below 50 % otherwise | Extend opening hours and add temporary seating only in those weeks | Lower cost and better service when it matters |
| Several sections use rooms with 120 seats for 35 students | Reassign sections to smaller rooms | Large halls become available for large classes |

In both applications the **finding** is a measured pattern, the **decision** is a concrete action by a named person, and the **effect** can be measured again later. This closed loop is what makes the decision *informed*.

## TASK 3 – Identify the Facets of Data 5 Marks

The university collects examination marks, attendance details, library transactions, online feedback and learning-platform activity. Classify these examples according to their data form.

- Identify structured, semi-structured and unstructured data wherever applicable.
- Give one university-related example for each form.
- Explain the key characteristics that distinguish these forms of data.

### 1. Classification of the university's data

The three forms are told apart by **how much fixed structure (schema) the data has**. Structured data fits fixed rows and columns, semi-structured data carries its own tags or keys but its fields can vary, and unstructured data has no predefined layout. Table 6 classifies the five examples given in the question. Some sources contain more than one form, so the table shows the main form and the other forms present.

**Table 6: Classification of the university's data by form**

| Data in the university | Typical content | Main form | Why | Other forms present |
|---|---|---|---|---|
| Examination marks | student ID, course code, test, marks, grade | **Structured** | Fixed fields, numeric and text values, stored in relational tables | Scanned answer sheets (unstructured) |
| Attendance details | student ID, course, date, period, present / absent | **Structured** | Same fields for every record; one row per student per period | Raw device logs from RFID or biometric readers (semi-structured) |
| Library transactions | student ID, book ID, issue date, due date, return date | **Structured** | Relational loan table | Search-and-click logs of the online catalogue (semi-structured) |

### 2. One university example for each form

#### Structured data – examination marks

**Table 7: Structured data: rows and columns with a fixed schema**

| student_id | course_code | test | marks (out of 50) | grade |
|---|---|---|---|---|
| S1001 | DSA04 | Internal 1 | 41 | A |
| S1002 | DSA04 | Internal 1 | 38 | B |
| S1003 | DSA04 | Internal 1 | 27 | C |

**Semi-structured data – learning-platform activity (JSON event)**

```
{ "user": "S1001", "time": "2026-09-14T21:42:10",
  "event": "video_play", "course": "DSA04",
  "details": { "video_id": "V12", "seconds_watched": 340 } }
{ "user": "S1002", "time": "2026-09-14T21:45:03",
  "event": "quiz_attempt", "course": "DSA04", "score": 7 }
```

Both records carry their own labels, but the second has a *score* and the first has a *details* block. A fixed table would need many empty columns, which is why this form is kept as JSON or in a NoSQL store.

### 3. Characteristics that distinguish the three forms

**Table 8: Key characteristics of the three data forms**

| Characteristic | Structured | Semi-structured | Unstructured |
|---|---|---|---|
| **Schema** | Fixed, defined before data is stored | Flexible; each record carries its own tags or keys | None |
| **Organisation** | Rows and columns in tables | Hierarchy of key–value pairs or tags (JSON, XML, CSV logs) | Raw text, images, audio, video, documents |
| **Effort to analyse** | Lowest | Medium | Highest |
| **Flexibility** | Low – a change to the schema is costly | High | Very high |
| **University example** | Marks, attendance, loans | LMS events, access logs, feedback export | Comments, forum posts, lecture videos, scanned papers |

### 4. Other facets of data present in the scenario

Data can also be described by how it is produced or what it contains. Several of these facets appear in the university's data and affect how it must be handled:

**Table 9: Other facets of data in the Smart University**

| Facet | University example | Handling implication |
|---|---|---|
| **Natural language** | Feedback comments, forum posts, emails to the helpdesk | Text analysis for topics and sentiment |
| **Machine-generated** | LMS click logs, access-card swipes, server logs | Large and fast; parsed automatically |
| **Streaming** | Live quiz events, real-time attendance scans | Processed as they arrive, not in nightly batches |
| **Graph / network** | Which students collaborate in forums and group projects | Stored and analysed as connections between nodes |
| **Audio, video and image** | Recorded lectures, scanned answer scripts, ID photos | Largest storage need; special processing |

**Conclusion.** The university's data is not of one form. Marks, attendance and loans are **structured**; platform activity and logs are **semi-structured**; comments and media are **unstructured**. Correct classification matters because the form decides **where the data is stored, which tools read it and how much preparation it needs** before analysis.

## TASK 4 – Analyse the Characteristics of Big Data 5 Marks

Analyse how the characteristics of Big Data may appear in the university's growing digital information.

- Explain the Volume, Velocity and Variety characteristics of Big Data.
- Relate each characteristic to a suitable university example.
- Analyse one challenge that may arise from each characteristic.

**Evidence Expected**
Correct explanation and scenario-based analysis of the Big Data characteristics.

### 1. The three characteristics at a glance

Big Data is data that is too large, too fast or too varied for a single traditional database and ordinary tools to store and process well. These three properties were first described together as the *3 Vs* [2]. Table 10 explains each one and links it to the university.

**Table 10: Volume, Velocity and Variety in the university**

| Characteristic | Meaning | University example | Main challenge |
|---|---|---|---|
| **Volume** | The amount of data stored | LMS events and recorded lectures: about 29 GB a day, over 10 TB a year | Storage and processing outgrow one server |
| **Velocity** | The speed at which data is produced and must be handled | Hundreds of LMS events every second during an online test | Alerts arrive too late if data is processed only once a night |
| **Variety** | The different forms and formats of data | Tables, JSON logs, comments, PDFs and videos | Data cannot be joined or analysed until formats are made compatible |

### 2. Volume

**Explanation.** Volume is the sheer quantity of data. In the university it grows every day because every student produces records in several systems, and nothing is deleted at the end of the semester.

**Table 11: Estimated daily data volume (based on the stated assumptions)**

| Source | Calculation | Daily volume |
|---|---|---|
| Attendance | 15,000 students × 6 periods × about 0.2 KB | about 18 MB |
| LMS click events | 15,000 students × 150 events × 0.5 KB | about 1.1 GB (2.25 million events) |
| Recorded lectures | 40 hours × 0.7 GB per hour | about 28 GB |
| Exam, library, feedback records | Small tables; grow by semester | below 1 GB per day on average |
| **Total** | | **about 29 GB a day, about 10.6 TB a year** |

**Analysis of the challenge.** A single database server and spreadsheet software cannot store and query several terabytes economically; reports that took seconds start to take hours, and backup and storage cost rise every year.

### 3. Velocity

**Explanation.** Velocity is how fast data arrives and how fast it must be used. Most university data arrives steadily, but there are sharp peaks.

**University example.** During an online test, if 3,000 students are connected and each produces one event (answer saved, page change) about every 5 seconds, the platform receives about **600 events per second**. On the day results are published, thousands of students log in within minutes.

### 4. Variety

**Explanation.** Variety is the number of different forms, formats and structures in the data.

**University example.** The same student appears as a row in the SIS table, a daily attendance record, a score in a marks file, a JSON event in the LMS log, a scanned answer script, and a comment in the feedback system. The student is called a *roll number* in one system and a *register number* in another, and dates are written as 14-09-2026 in one and 2026-09-14 in another.

**Analysis of the challenge.** Each form needs its own storage and tools, and **the sources cannot be combined until a common key, date format and structure exist**. Free text has to be converted into numbers

**Other characteristics.** *Veracity* (how trustworthy the data is) is often added as a fourth V. In the university it appears as wrong or incomplete records, and it is handled by the cleansing stage of Task 6. It also depends on the other three: more volume, speed and variety make quality harder to control.

## TASK 5 – Big Data Ecosystem and Its Components 5 Marks

Explain the role of a Big Data ecosystem in managing information collected from multiple university systems.

- Identify the major stages/components involved in data collection, storage, processing and analysis.
- Explain the purpose of each stage/component.
- Draw a labelled block diagram showing how data can move through the ecosystem.
- Analyse how the ecosystem helps the university handle large and varied datasets.

**Evidence Expected**

A labelled and logically connected ecosystem diagram with explanations of its components and relevance.

### 1. Role of the Big Data ecosystem

A Big Data ecosystem is a **set of connected technologies, each doing one job**, that together collect, store, process and analyse data that is too big, fast or varied for one machine or one database. For the university it is the platform that sits between the six source systems and the people who use the findings. Its role is to bring the data of all systems into one managed environment, keep it safe, prepare it, and make it available for analysis and presentation – without the source systems (such as the live examination system) being slowed down by analysis work.

### 2. Components and their purpose

Table 12 lists the stages that the data passes through, the purpose of each, examples of technology that provide it, and its use in the university. The numbers match the labelled block diagram in Figure 2.

**Table 12: Components of the Big Data ecosystem and their purpose**

| Stage / component | Purpose | Example technology | University use |
|---|---|---|---|
| **1. Data sources** | Where data is generated | SIS, attendance, exam, LMS, library, feedback systems | Structured tables, JSON logs and text comments |
| **2. Data ingestion (collection)** | Moves data from the sources into the ecosystem; **batch** loads for scheduled data and **streaming** for live events | ETL tools, Sqoop-type loaders; Kafka-type message queues | Nightly load of marks and library records; live stream of LMS and attendance events |

| Stage / component | Purpose | Example technology | University use |
|---|---|---|---|
| **3. Data storage** | Keeps data safely and cheaply in a form suited to its use | Distributed file system or object store (data lake); data warehouse; NoSQL store | Raw logs and text in the lake; cleaned tables in the warehouse; JSON events in NoSQL |
| **4. Data processing** | Cleans, joins and summarises very large data by splitting the work across many machines | MapReduce, Spark (batch); Spark Streaming, Flink (stream) | Compute weekly attendance and logins for 15,000 students; raise live alerts |
| **6. Presentation / use** | Delivers findings in a form that users can understand | BI dashboards, reports, alerts | Principal, HoDs, coordinators and mentors |
| **Resource management and scheduling** | Allocates machines and runs jobs in the right order | Cluster manager, workflow scheduler | Runs the nightly load, then cleansing, then analysis |
| **Security, privacy and governance** | Controls who can see data; protects personal data; records where data came from | Access control, encryption, anonymisation, metadata catalogue | Coded student keys; role-based access; audit log |

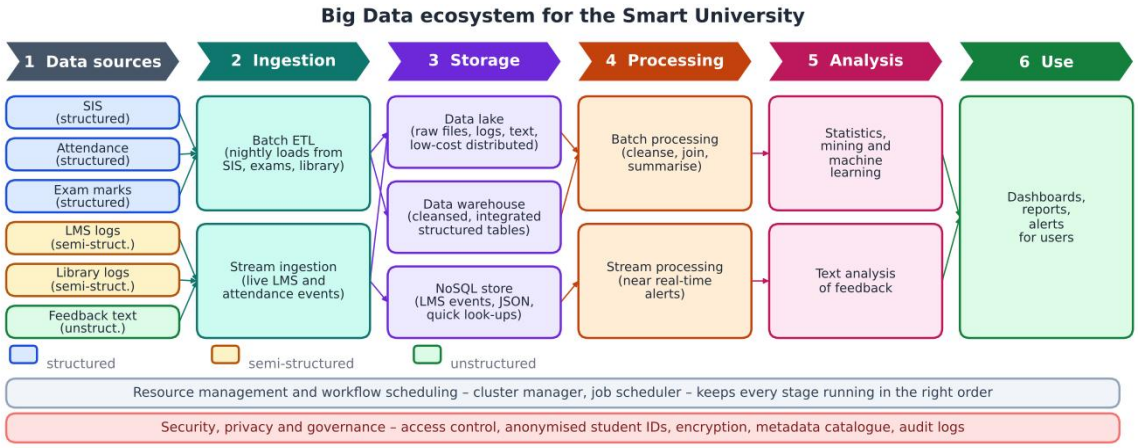



*Figure 2: Labelled block diagram of the Big Data ecosystem – flow of data from the university systems to the users*

### 3. How data moves through the ecosystem

Following one LMS event shows how the stages connect:

1. A student plays a lecture video. The LMS writes a JSON event (**stage 1**).

2. The event is sent to the **stream ingestion** queue within seconds; the nightly **batch load** brings in the marks and library tables (**stage 2**).
3. The raw event is stored in the **data lake**, and a copy goes to the **NoSQL store** for quick look-up. Marks and attendance wait in the **warehouse** (**stage 3**).
4. **Batch processing** cleans and joins the data and counts logins per student per week. **Stream processing** checks live data against alert rules (**stage 4**)

**4. How the ecosystem helps with large and varied datasets**

**Table 13: Contribution of the ecosystem to handling large and varied datasets**

| **Problem in the university** | **How the ecosystem helps** |
|---|---|
| **Volume** grows to over 10 TB a year | Storage and processing are **spread over many machines**; capacity is increased by adding machines instead of replacing one large server |
| **Velocity** – bursts of events during tests and results | A **buffered streaming path** takes the peak load and a stream processor reacts to events at once, while heavy work continues in batch |
| **Variety** – tables, JSON, text and video | The **data lake accepts any format** as it arrives; structure is added when needed, and each data form is kept in the store best suited to it |
| Separate departments, separate systems | All data reaches **one governed environment**, so a student can be viewed across all six systems |
| Hardware failure | Data is **replicated** across machines, so one failure does not lose records or stop analysis |
| Analysis slowing down live systems | Analysis runs on the copy in the ecosystem; the **source systems are not disturbed** |

**Analysis.** The ecosystem does not make the data better by itself; it makes it **possible to handle at scale**. Its benefits come with costs: more components to maintain, a need for skilled staff, and a stronger duty to protect personal data. For a university of this size the volume of video and the need for early alerts justify these costs, but the ecosystem can be started small (one lake, one warehouse, one scheduler) and expanded when the data grows.

## TASK 6 – Data Retrieval and Data Cleansing 4 Marks

The university plans to combine student information, attendance and examination records. Explain how data retrieval and cleansing should be carried out before analysis.

- Identify suitable data sources and the information to be retrieved.
- Explain how missing, duplicate, inconsistent or incorrect records can be identified and handled.
- State why data quality is important for reliable decision-making.

**Evidence Expected**

A suitable retrieval plan and clear explanation of cleansing activities and their importance.

### 1. Where retrieval and cleansing fit in the process

The Data Science process has stages that follow one another, with repeats whenever the results are weak [1]. Retrieval and cleansing are the second and third stages, and they decide what everything after them is built on.




*Figure 3: Stages of the Data Science process used in this report*

### 2. Retrieval plan

Retrieval starts with a clear question: *which students, in which courses and in which period?* Only the data needed for that question is retrieved, and personal details that are not needed (name, address, phone number) are left behind.

**Table 14: Retrieval plan for student, attendance and examination data**

| Source | Information to be retrieved | Key field | How it is retrieved |
|---|---|---|---|
| Student information system | student ID, programme, year, section, admission category | student ID | Scheduled database query or export (batch) |
| Attendance records | student ID, course code, date, period, present / absent | student ID + course + date | Daily extract; live feed if alerts are needed |
| Examination system | student ID, course code, assessment, maximum marks, marks obtained, grade | student ID + course + assessment | Extract after each internal test and end-semester result |

Retrieval rules: use **read-only access** so the source system is never changed; retrieve **the same period** from all three sources; record the **date and time of each extract** and the number of rows received, and compare the row count with the source to be sure that nothing was lost in transfer; store the untouched copy in the raw zone before any cleaning, so that every correction can be traced.

### 3. Cleansing: finding and handling bad records

Cleansing means finding records that are missing, repeated, contradictory or wrong, and correcting or removing them by clear rules. Table 15 shows how each type is found and handled.

**Table 15: Identifying and handling data-quality problems**

| Problem | How it is identified | How it is handled | Example |
|---|---|---|---|
| **Missing** | Count empty or NULL values per column; compare expected and actual number of records (e.g., 60 periods but only 52 attendance rows) | Find the reason first. Fill from another source if it exists; otherwise keep it as *missing* and exclude it from that calculation. Do **not** replace marks with 0, because absent is not the same as zero. Use a mean only for non-critical numeric fields, and note it | Internal marks of S1003 not yet entered |
| **Duplicate** | Check for repeated keys (student ID + course + date; student ID in the SIS table); compare full rows | Keep one record (the latest or the most complete) and delete the rest; log the number removed | S1001 appears twice because the record was loaded twice |

**Worked example (illustrative)**

**Table 16: Raw extract before cleansing**

| student_id | programme | attendance % | internal marks (/50) |
|---|---|---|---|
| S1001 | CSE | 92 | 41 |
| S1001 | CSE | 92 | 41 |
| S1002 | C.S.E. | 105 | 38 |
| S1003 | Computer Science | 78 | (blank) |
| S1004 | CSE | 64 | 54 |

**Table 17: The same extract after cleansing**

| student_id | programme | attendance % | internal marks (/50) | Action taken |
|---|---|---|---|---|
| S1001 | CSE | 92 | 41 | Duplicate removed |
| S1002 | CSE | 85 | 38 | Label standardised; attendance recalculated from the daily register (105 was impossible) |
| S1003 | CSE | 78 | missing | Kept as missing, not 0; excluded from marks analysis; sent for entry |
| S1004 | CSE | 64 | 45 | Corrected from the mark sheet (54 was a typing error, above the maximum of 50) |

**Cleansing steps in order**

1. Profile the raw data (counts, distinct values, minimum and maximum of each field).
2. Remove exact duplicates and resolve repeated keys.
3. Standardise formats, spellings and units.
4. Validate values against range and business rules.
5. Handle missing values by source, rule or exclusion.
6. Re-run the checks and compare counts; keep a **cleansing log** of every rule and the number of records changed.

**4. Why data quality is important**

- **Wrong data gives wrong decisions.** If attendance is overstated, an at-risk student is missed; if marks are replaced with 0, a good student is wrongly flagged. In both cases support goes to the wrong people.
- **Errors multiply when data is combined.** A duplicate student record in one source can become several rows after joining and distort every average.

## TASK 7 – Data Integration and Transformation 5 Marks

Explain how information obtained from different university systems can be integrated and transformed into a suitable dataset for analysis.

- Describe the purpose of data integration.
- Explain the need for data transformation using a relevant example.
- Identify possible issues when combining data with different formats or structures.

**Evidence Expected**

A logical explanation of integration and transformation, supported by relevant examples.

### 1. Purpose of data integration

Data integration **combines data from different systems into one consistent dataset** so that each student can be studied as a whole. Alone, the marks show *what* a student scored, the attendance shows *how often* the student came, and the LMS log shows *how the student studied*; the reasons for a result appear only when these are placed side by side. Integration also removes the need to compare several reports by hand and gives every user the same version of the facts.

### 2. How the sources are integrated

1. **Choose the unit of analysis** – here, one row per student per semester.
2. **Choose a common key.** The student ID is used. Where a system uses a different identifier (register number, e-mail), a **mapping table** converts it to the student ID.
3. **Standardise** formats, date styles, units and code lists in every source (as in Task 6).
4. **Summarise** each many-row source to one row per student **before** joining: for example, 60 daily attendance rows become one attendance percentage. This prevents the join from multiplying rows.
5. **Join** the summarised tables on the key. A **left join from the student table** keeps every student, even one with no LMS activity, whose absence is itself a finding.
6. **Check the result**: the number of rows equals the number of students, no student appears twice, and a few students are verified by hand against the sources.

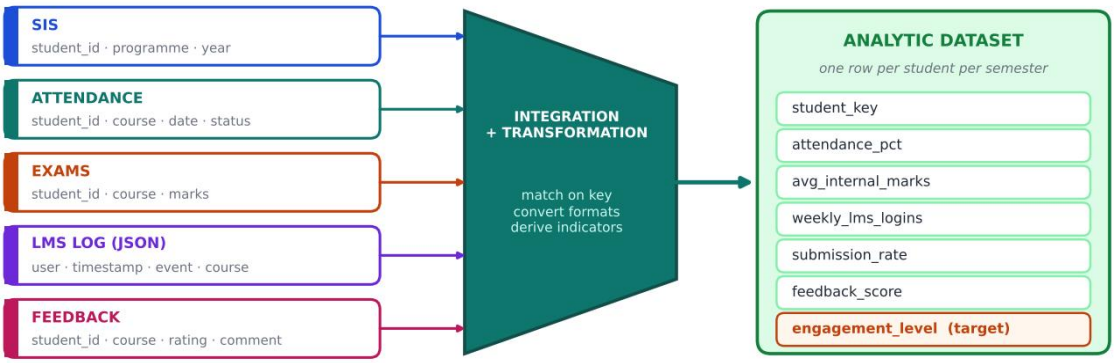



*Figure 4: Integration and transformation of five source systems into one analytic dataset*

### 3. Need for data transformation – with an example

Transformation **changes the form or scale of data so that it can be compared and analysed**. Raw data is recorded for the convenience of each source system, not for analysis. For example, the attendance system stores one row for each student for each period; a model cannot use 60 separate rows, but it can use *attendance percentage*. Likewise, marks out of 50 in one test and out of 100 in another cannot be averaged until they are converted to the same scale.

**Table 19: Transformation example for one student, S1001 (illustrative)**

| Raw data | Transformation | Result in the analytic dataset |
|---|---|---|
| 60 attendance rows, 54 marked *present* | Count and divide: 54 ÷ 60 × 100 | attendance_pct = 90.0 |
| Internal marks 41 out of 50; test 2: 70 out of 100 | Convert to percentage, then average: 82 % and 70 % | avg_internal_pct = 76.0 |
| LMS log: 18 login events in 6 weeks | Count per week: 18 ÷ 6 | weekly_logins = 3.0 |
| 8 assignments set, 7 submitted | 7 ÷ 8 × 100 | submission_rate = 87.5 |
| Comment: "videos too long, quiz page fails at night" | Text analysis gives a sentiment score from −1 to +1 | feedback_score = −0.4 |
| Programme names, grades (A, B, C) | Coding: standard programme code; grade converted to points | programme = CSE; grade_points = 9 |
| weekly_logins ranges from 0 to 12 across students | Min–max scaling: (3 − 0) ÷ (12 − 0) | logins_scaled = 0.25 |

Scaling to 0–1 is needed because a method that measures similarity between students would otherwise be dominated by the field with the largest numbers (marks up to 100) over a field with small numbers (logins per week), although both matter equally.

### 4. Issues when combining data of different formats or structures

**Table 20: Issues in integrating data of different formats and structures**

| Issue | Example in the university | How it is handled |
|---|---|---|
| **Different keys** | Roll number in attendance, register number in exams | Mapping table to the student ID; check unmatched keys |
| **Different levels of detail** | Attendance daily, marks per test, feedback per semester | Summarise to the unit of analysis (student per semester) |
| **Different structures** | Tables (SIS) and nested JSON (LMS) | Flatten the JSON to tables; keep the key fields |

| Issue | Example in the university | How it is handled |
|---|---|---|
| **Unstructured data** | Feedback comments | Convert to numbers (topic, sentiment) first; keep the original text |
| **Duplicates created by joins** | Joining unsummarised daily rows with marks | Summarise first; test that the row count is unchanged |
| **Different meanings of a term** | "Attendance" in the exam system means eligibility, in the attendance system presence | Agree and document one definition in a data dictionary |
| **Privacy** | Combining sources makes a student identifiable | Replace names with a coded key; restrict access |

**Outcome.** The result is an **analytic dataset** with one clean, numeric, comparable row per student, from which the analysis in Task 8 can start. The rules used for every conversion are stored with the dataset so that the same steps can be repeated each week.

## TASK 8 – Data Analysis and Model-Building Stages 5 Marks

After preparing the university dataset, explain the data analysis and model-building stages in the Data Science process.

- Describe the purpose of analysing the prepared data.
- Explain, at a conceptual level, how a model may be built to identify patterns in student engagement.
- State why the analysis and model-building stages should be based on the quality and relevance of the data.

**Evidence Expected**

A clear description of analysis and model-building stages connected to the scenario. No implementation of a machine-learning algorithm is required.

**Ans:**

### 1. Purpose of analysing the prepared data

Analysis (also called exploratory analysis) is the first look at the prepared dataset to **understand it before any model is built**. Its purposes are to describe what the data shows, to check that it makes sense, to find relationships and unusual cases, and to decide which indicators are worth modelling. For the university it answers questions such as those in Table 21.

**Table 21: Data-analysis activities for the student-engagement dataset**

| Analysis activity | Question for the university | Possible output |
|---|---|---|
| **Summary statistics** | What are the average and spread of attendance, marks and weekly logins? | Mean, median, minimum, maximum per course |
| **Distribution** | Do most students cluster near the average or are there two distinct groups? | Histogram of weekly logins |
| **Comparison of groups** | Does engagement differ between sections, years or programmes? | Bar chart by section |
| **Relationship between indicators** | Do students with low attendance also have low LMS activity and low marks? | Correlation table or scatter plot |
| **Outliers** | Which students are very different from the others? | List for checking, as an error or a real case |

Analysis also checks the data itself. If the average attendance of a section is exactly 100 %, or a column is empty for a whole programme, this indicates a data problem that must go back to the cleansing stage, not a result to report.

**2. Building a model to identify engagement patterns (conceptual)**

A **model** is a simplified description of the pattern in the data, which can be applied to students not seen before. The aim here is to identify *engagement patterns* – groups of students who behave in a similar way – and, in addition, to spot students moving towards low engagement. The stages are:

1. **State the modelling goal.** For example: "Group students by their engagement behaviour in the semester" and "identify students whose engagement is likely to be low."
2. **Select features** that describe engagement and are relevant to the goal: attendance percentage, weekly LMS logins, assignment submission rate, quiz participation, library visits and the change in these values over recent weeks. Fields unrelated to behaviour (such as the student's hostel room number) are left out.
3. **Prepare the data for modelling.** Use the scaled values from Task 7. Keep part of the data aside (for example, one previous semester) to test the model later.
4. **Validate.** Check the result on data not used for fitting: are the groups stable, do they make sense to the faculty, and does the classification find the students who really became low-engagement? Check for wrong predictions of both kinds (missing a student at risk, or flagging a student who was fine).
5. **Interpret and name the groups,** and improve the model by returning to earlier stages (add features, correct data) if it does not make sense.

**Table 22: Illustrative result of grouping students by engagement (values assumed)**

| Group (named by the team) | Typical attendance | Typical weekly logins | Typical submission rate | Interpretation |
|---|---|---|---|---|
| Highly engaged | about 90 % | about 8 | about 95 % | Regular, independent learners |
| Moderately engaged | about 78 % | about 4 | about 75 % | Study mostly before deadlines; may slip |
| Low engaged | about 58 % | about 1 | about 40 % | At risk; need early support |

The model itself does not give names or reasons; it only places students with similar behaviour together. The **meaning** of each group is decided by people who study the typical values (as in Table 22) and use their knowledge of the courses. A student's group is then reviewed every week, so that a move from *moderate* to *low* engagement creates an alert.

**3. Why quality and relevance of the data matter**

A model can find only the patterns that are present in the data it is given, and it cannot judge whether the data is right. Therefore the analysis and model-building stages depend completely on what is prepared before them:

**Table 23: Effect of data weaknesses on analysis and model building**

| Weakness in the data | Effect on analysis and model |
|---|---|
| Wrong or missing values (poor quality) | Wrong averages and groups; students flagged or missed by mistake |
| Features that are irrelevant to engagement | Noise hides the real pattern; groups are meaningless |
| Features that already contain the answer (e.g., using final marks to predict who will fail) | The model looks accurate in testing but is useless in practice because the information is not available early enough |
| Data from only one course, one semester or one section | Findings do not hold for the whole university |
| Biased records (e.g., attendance recorded differently by different faculty) | Unfair comparison of sections; unfair decisions about students |
| Old data that no longer describes present behaviour (e.g., before online learning increased) | The model describes students who no longer exist |

**Therefore** the team checks quality before analysing, chooses features for their relevance to engagement, tests on separate data, and repeats the stages whenever the result is not credible.

## TASK 9 – Presenting Findings and Building Applications 5 Marks

The university management wants to understand the results of the student-engagement analysis. Propose how the findings can be presented through a simple decision-support application.

- Identify the intended users of the application.
- Suggest suitable presentation methods such as summary tables, charts or dashboards.
- Specify at least three useful indicators that may be displayed.
- Explain how the application can help administrators make decisions.

**Evidence Expected**

A relevant application concept, suitable presentation methods and explanation of decision-making value.

**Ans:**

### 1. Application concept

I propose a web-based **Student Engagement Decision-Support Dashboard**. It reads the weekly analytic dataset and the engagement groups, and shows the results in a few pages. The page is updated automatically every night (and in near real time for alerts), so users always see the present position. Each user sees only what is relevant to the role, and student names appear only for those who must act on them.

### 2. Intended users

**Table 24: Intended users of the application**

| User | Main question | What they see |
|---|---|---|
| **Principal / Dean** | How engaged are our students overall and is it improving? | University summary, trends by department |
| **Head of Department** | Which courses and sections need attention? | Comparison of courses and sections; list of alerts |
| **Course faculty / coordinator** | How are my students responding to my course? | Course indicators, weekly trend, at-risk list for the course |
| **Mentors and student counsellors** | Whom should I meet this week? | Named list of students flagged, with the reason |
| **Academic office / planning** | Where do we need to change timetable, rooms or support? | Utilisation and engagement summaries |

### 3. Presentation methods

**Table 25: Suitable presentation methods**

| Method | Best used for | Example in the application |
|---|---|---|

| Method | Best used for | Example in the application |
|---|---|---|
| **Summary table** | Exact figures; comparing many courses | Table of courses with attendance, logins and at-risk % |
| **KPI cards** | The few numbers management checks first | Average attendance, active LMS users, students at risk |
| **Line chart** | Change over time | Weekly LMS participation against the target |
| **Bar chart** | Comparing groups | At-risk percentage by course or section |
| **Heat map** | Patterns across two dimensions | Engagement by section and week |
| **Alert list with drill-down** | Prompting action; going from summary to detail | Click a course to see its sections, then its flagged students |
| **Dashboard** | Several views on one screen, with filters | Figure 5 |

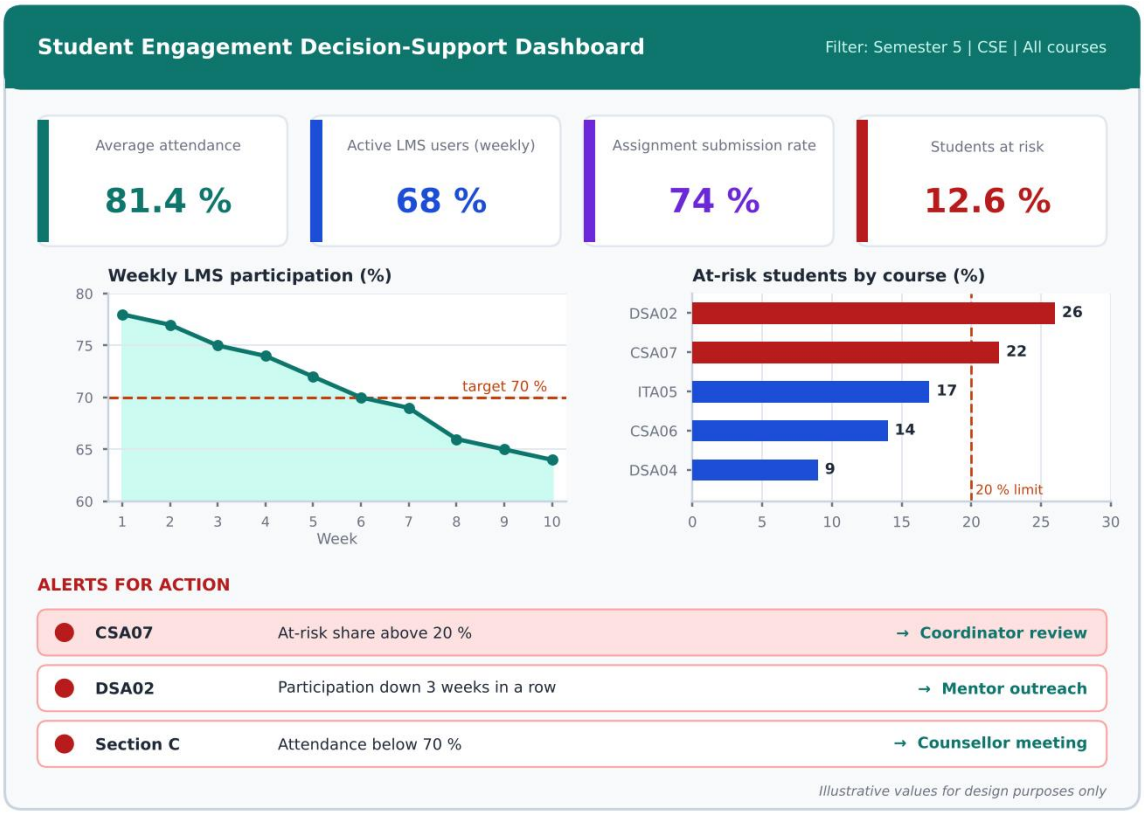



*Figure 5: Outline (wireframe) of the proposed dashboard – all values are illustrative*

## 4. Indicators to be displayed

**Table 26: Indicators shown by the application**

| Indicator | Definition | Use in decisions |
|---|---|---|
| **Average attendance %** | Periods attended ÷ periods held × 100, per course or section | Below 75 % triggers a review (the usual eligibility limit) |
| **Weekly active LMS users %** | Students with at least one LMS activity in the week ÷ enrolled students | A fall for 3 weeks in a row triggers mentor outreach |
| **Students at risk %** | Students in the low-engagement group ÷ all students | Above 20 % in a course triggers coordinator review |
| **Average internal marks** | Mean of internal-test marks (as %) | Compared with the previous semester |

## 5. How the application helps administrators decide

**Table 27: From the dashboard to decisions**

| What the administrator sees | Decision taken | Who acts |
|---|---|---|
| Students at risk is 22 % in CSA07 against 9–14 % elsewhere | Review of the course (pace, assessment, extra tutorials) | Head of Department and coordinator |
| Feedback score is low and comments mention long videos | Shorten videos; add quizzes | Course faculty |
| Laboratory and library use are low at certain times | Rearrange sessions and opening hours | Academic office; librarian |

The application helps because it **reduces the time between a problem and the action**: instead of waiting for a report, administrators see the warning with a reason and an owner. It makes decisions **consistent** (the same indicator and limit for every course), **traceable** (the data behind each number can be opened) and **checkable** (the same indicators show whether the action worked).

### Design notes and assumptions

- Data is refreshed nightly; alerts are refreshed during the day.
- Role-based login; student names visible only to mentors and counsellors; all views are logged.
- The thresholds (75 %, 20 %, three weeks) are starting values set by the university and reviewed each semester.
- The application **supports** human decisions; it does not decide about any student automatically.

## TASK 10 – Integrated Data Science Decision-Making 8 Marks

Suppose the university observes a decline in student participation in online learning activities. Analyse how the complete Data Science process and the Big Data ecosystem can help investigate the issue and support an informed decision.

- Identify relevant data sources and the facets/forms of data involved.
- Outline the steps required for retrieval, cleansing, integration, transformation and analysis.
- Explain how findings can be presented to decision-makers.
- Recommend two possible data-informed actions and justify how the analysis could support them.
- Represent the overall approach using a suitable flowchart.

**Evidence Expected**

An integrated, logically ordered analysis connecting data forms, the Big Data ecosystem, the Data Science process, presentation of findings and justified decisions.

**Table 28: Data sources and data forms for the investigation**

| Source | Data used for this problem | Form of data |
|---|---|---|
| **LMS activity log** | Login, video play, quiz attempt, forum post, with time stamp and course | **Semi-structured** (JSON events); **machine-generated, streaming** |
| **LMS content and forum** | Discussion posts, uploaded files, lecture videos | **Unstructured** |
| **Attendance records** | Class attendance by student, course and date | **Structured** |
| **Student information system** | Programme, year, section, hosteller or day scholar | **Structured** |
| **Examination system** | Internal marks and trend | **Structured** |
| **Online feedback platform** | Ratings and written comments on the courses | **Structured** ratings and **unstructured** text |

### 3. Role of the Big Data ecosystem

**Table 29: Use of the Big Data ecosystem in the investigation**

| Ecosystem component | Contribution to this investigation |
|---|---|
| **Ingestion** | Streams the LMS events and loads the other systems in nightly batches, so that all ten weeks of data are available |
| **Storage** | Keeps raw logs, comments and files in the data lake; cleaned tables in the warehouse; recent events in the NoSQL store for quick checks |
| **Processing** | Handles the millions of events (about 2.25 million per day) by splitting the work over many machines; summarises them to weekly counts per student |
| **Analysis** | Compares groups, finds relationships and groups students; analyses feedback text |

### 4. Steps of the Data Science process for this problem

1. **Retrieval.** Retrieve ten weeks of LMS events, attendance, internal marks, student profile, feedback and LMS server logs for all courses of the semester, with the extract time and row counts recorded. Keep the raw data unchanged in the lake.
2. **Cleansing.** Remove duplicate events caused by page reloads; standardise course codes and time zone; find students with no student-profile match; check impossible values (activity time before the semester starts, 24-hour video watch time); treat missing marks as *missing*, not zero; record all changes in a cleansing log.
3. **Integration.** Summarise events to one row per student per week; join with attendance, marks, profile and feedback on the student key (left join from the profile table); verify that the number of students is unchanged.
4. **Transformation.** Derive weekly logins, videos watched, quizzes attempted, submission rate and attendance percentage; convert marks to percentage; convert comments to topic and sentiment scores; scale features to a 0–1 range.
5. **Analysis.** Study the participation trend and compare it between groups, as in Table 30. Then group students by engagement to find the part of the student body that changed.

**Table 30: Analysis questions for the decline in participation**

| Analysis question | Method | Example of a possible finding (assumed) |
|---|---|---|
| When did participation start to fall? | Weekly trend line | Fall began in week 4 and became steeper after the first internal tests |
| Where is the fall concentrated? | Compare by course, section, year | Most of the fall is in two courses (CSA07, DSA02) and in year 2 |
| Who has stopped participating? | Group by engagement; compare with attendance and marks | The fall is mainly in the group with attendance below 75 % |
| At what time are students active? | Activity by hour and day | Evening activity (9–11 pm) is highest but failed-login rate is also highest then |

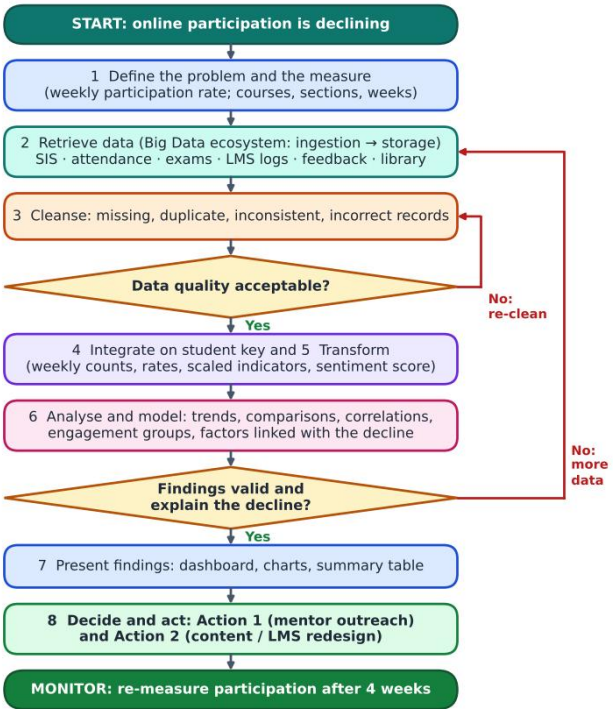



*Figure 6: Flowchart of the integrated Data Science approach to the decline in online participation*

The flowchart shows two repeat paths. If the data is not of acceptable quality, the team returns to cleansing. If the findings do not explain the decline, the team goes back to retrieve more data or to derive more features (for example the time of day). The decision is not taken until both checks are passed.

## 5. Presenting the findings to decision-makers

- **A one-page summary for the Principal and HoDs:** the headline (participation 78 % → 64 %), the main causes in plain language, and the two recommended actions.

- **Dashboard views:** weekly participation line chart against the 70 % target; bar chart by course; heat map by section and week; table of courses with their feedback topics.
- **Drill-down for coordinators and mentors:** the list of students in the low-engagement group and the reason flagged (shown only to those who must act).
- Findings are stated with their **limits**: which data was used, how complete it was, and which findings are associations, not proven causes.

### 6. Two data-informed actions and their justification

#### Action 1 – Targeted mentor outreach for the students who have stopped participating

**What.** Mentors and counsellors contact the students in the low-engagement group in the two most affected courses, find the reasons (workload, connectivity, loss of interest, personal reasons) and agree a plan with each student.

**How the analysis supports it.** The grouping showed that the decline is concentrated in a defined group (low attendance, low logins) and not spread evenly; so a *targeted* action reaches the students who need it, at far lower effort than a campaign for everyone. The early-alert rule (participation down three weeks in a row) allows the mentors to act in the week the pattern appears. **Measure of success:** the participation rate of the contacted group after four weeks, compared with similar students who were not contacted.

#### Action 2 – Redesign of course content and correction of LMS access problems

**What.** In the courses highlighted by the feedback text and the server logs, divide long videos into short segments, add a short weekly quiz with feedback, and correct the night-time failures of the quiz page (for example by increasing the capacity at peak hours).

## Overall Conclusion

This report applied the CO1 concepts to the Smart University scenario. Data Science is required because the university's data is spread across six systems, large, fast and varied, while its decisions must be early and evidence-based (Tasks 1–2). The data takes structured, semi-structured and unstructured forms (Task 3) and shows the Volume, Velocity and Variety of Big Data with a challenge in each (Task 4). A Big Data ecosystem of ingestion, storage, processing, analysis and presentation, with scheduling and security, makes it possible to manage this data (Task 5). The Data Science process then turns it into decisions: data is retrieved and cleansed (Task 6), integrated and transformed (Task 7), analysed and modelled (Task 8) and presented through a decision-support application (Task 9). Task 10 showed all of these working together on a real question – a decline in online participation – and ended with two justified, measurable actions.

Throughout, data quality and relevance and responsible use of student data were the conditions on which the reliability of every finding depends.